

SMART INDIA HACKATHON (SIH) 2026

Farmer-to-Buyer Marketplace & Supply-Chain Platform

Comprehensive Codebase Diagnostic, Maturity Audit, Risk Evaluation, and Target Architecture Roadmap

PROBLEM STATEMENT	ACTIVE STACK	CURRENT MATURITY	TARGET EXECUTION
Direct Agrimarket & Logistics	React 19.2.8 + Vite 8.2.2	Landing Page & UI Shell (Phase 0)	AI-Enabled MVP (Phases 1-6)

1 Executive Summary

The goal of this SIH 2026 initiative is to build a transparent, AI-driven agricultural ecosystem that eliminates intermediary exploitation, maximizes farmer price realization, reduces end-consumer grocery bills, and streamlines farm-to-hub pickup and aggregated logistics.

A thorough, non-destructive audit of the existing repository confirms that the codebase currently functions as an **aesthetic, bilingual informational landing page** with custom responsive CSS tokens and modal triggers. However, **none of the functional backend, database, transaction, AI grading, or route optimization workflows are implemented yet.**

Core Assessment: The codebase is in a clean baseline state with an elegant design foundation and zero dependency bloat. With intentional architectural structuring (adding client routing, a dedicated AI backend service, and spatial database persistence), it is primed for rapid MVP development.

2 Repository Structure & File Inventory

The workspace contains 19 source assets under `src/`, 2 public files, and standard Vite config:

```
farmer-s-platform/
├─ package.json          # React 19.2.8, Vite 8.2.2, Oxlint
├─ vite.config.js        # Standard @vitejs/plugin-react
├─ index.html            # Single page root shell
├─ public/               # favicon.svg (Vite), icons.svg (unused)
└─ src/
  ├─ main.jsx            # React.StrictMode mount with index.css
  ├─ App.jsx             # Coordinator: LanguageProvider + HomePage
  ├─ index.css           # Core design system tokens & responsive layout
  ├─ App.css             # ⚠️ Orphaned Vite template CSS (185 lines, unimported)
  ├─ translations.js     # Complete bilingual dictionary (English / Hindi)
  ├─ context/
  │   └─ LanguageContext.jsx # React context providing { lang, toggleLang, t }
  ├─ components/
  │   ├─ Header.jsx      # Sticky navigation, brand, voice CTA, lang switch
  │   ├─ Hero.jsx        # Value proposition, farmer/buyer CTAs, custom SVG
  │   ├─ HowItWorks.jsx  # 7-step sequential workflow visualization
  │   ├─ Icons.jsx       # 10 hand-crafted vector SVG icon components
  │   ├─ Modal.jsx       # Reusable backdrop dialog with event stopPropagation
  │   ├─ UserTypes.jsx   # Persona dual-card selector (Farmer vs Buyer)
  │   └─ Footer.jsx      # SIH project attribution
  └─ assets/             # hero.png, react.svg, vite.svg (all unreferenced)
```

File / Component	Lines / Size	Architectural Responsibility	Audit Finding
src/context/LanguageContext.jsx	43 lines	Global language state & dictionary provider	FAST REFRESH WARNING Exports hook & component together
src/translations.js	169 lines	Static bilingual string mappings (EN / HI)	CLEAN STRUCTURE Outlines all 7 vision steps clearly
src/components/Modal.jsx	24 lines	Accessible popup dialog with click-outside dismiss	REUSABLE Clean accessible modal implementation
src/components/Icons.jsx	101 lines	Zero-dependency lightweight SVG React icons	OPTIMIZED Keeps bundle footprint minimal (no heavy icon library)
src/App.css	185 lines	Original Vite starter stylesheet	DEAD CODE Not imported anywhere in the project

3 Feature Maturity Audit: Implemented vs. Placeholders

Feature Component	Current State	Maturity	What Is Needed For MVP
Bilingual Switching (EN / HI)	Instant reactive translation toggle across all labels and modals	WORKING	Extend to dynamic backend data (crop names, notifications)
Visual 7-Step Supply Flow	Numbered, interconnected responsive workflow cards with icons	WORKING	Link each step card to the corresponding interactive portal
Farmer Listing & Onboarding	Clicking "I'm a Farmer" triggers an informational popup modal	PLACEHOLDER	Produce upload form: crop, variety, quantity, harvest date, GPS
Buyer Marketplace & Orders	Clicking "I'm a Buyer" triggers an informational popup modal	PLACEHOLDER	Product catalog, grade filtering, cart, order placement & escrow
AI Crop Quality Grading	Referenced in Step 2 of HowItWorks text	MISSING	Camera capture + Gemini Vision multimodal quality score (A/B/C)
AI Fair Price Recommendation	Referenced in Step 3 of HowItWorks text	MISSING	Mandi APMC live benchmark + quality grade multiplier formula
Smart Hub Aggregation	Referenced in Steps 5 & 6 of HowItWorks text	MISSING	Geographic clustering of farm coordinates to optimal pickup centers
Logistics & Route Optimization	Referenced in Step 7 of HowItWorks text	MISSING	Map visualization + Multi-stop TSP/VRP route planner (Leaflet)
Voice Assistant	Header microphone button shows "Coming soon" modal	PLACEHOLDER	Web Speech API speech-to-text parser for Hindi and English voice input

4 Architectural & Dependency Diagnostics

Runtime & Build System

React 19.2.8: Takes advantage of the latest React engine and high performance JSX runtime.

Vite 8.2.2: Ultra-fast hot module replacement, 282ms production build time, and Oxlint static analysis.

Styling: Pure CSS design tokens in index.css. No CSS-in-JS overhead; highly performant on low-end smartphones.

State & Data Management

Global State: Restricted to LanguageContext. No authentication state, user session, or cart storage currently exists.

Component State: Local state in HomePage controls modal visibility and selected user persona.

Network Layer: Zero HTTP client installed (no Axios or TanStack Query). All data is hardcoded in client memory.

5 Technical Debt & Pre-Integration Risks

Risk 1: Missing Git Version Control: The directory is not initialized as a git repository (fatal: not a git repository). Any destructive code modifications or refactoring cannot be rolled back without git tracking.

Risk 2: Oxlint Fast Refresh Penalty: In `LanguageContext.jsx`, exporting both `LanguageProvider` and `useLanguage()` causes Oxlint warning: `warning react(only-export-components)`. In Vite, this triggers full browser reloads during development instead of seamless hot module state preservation.

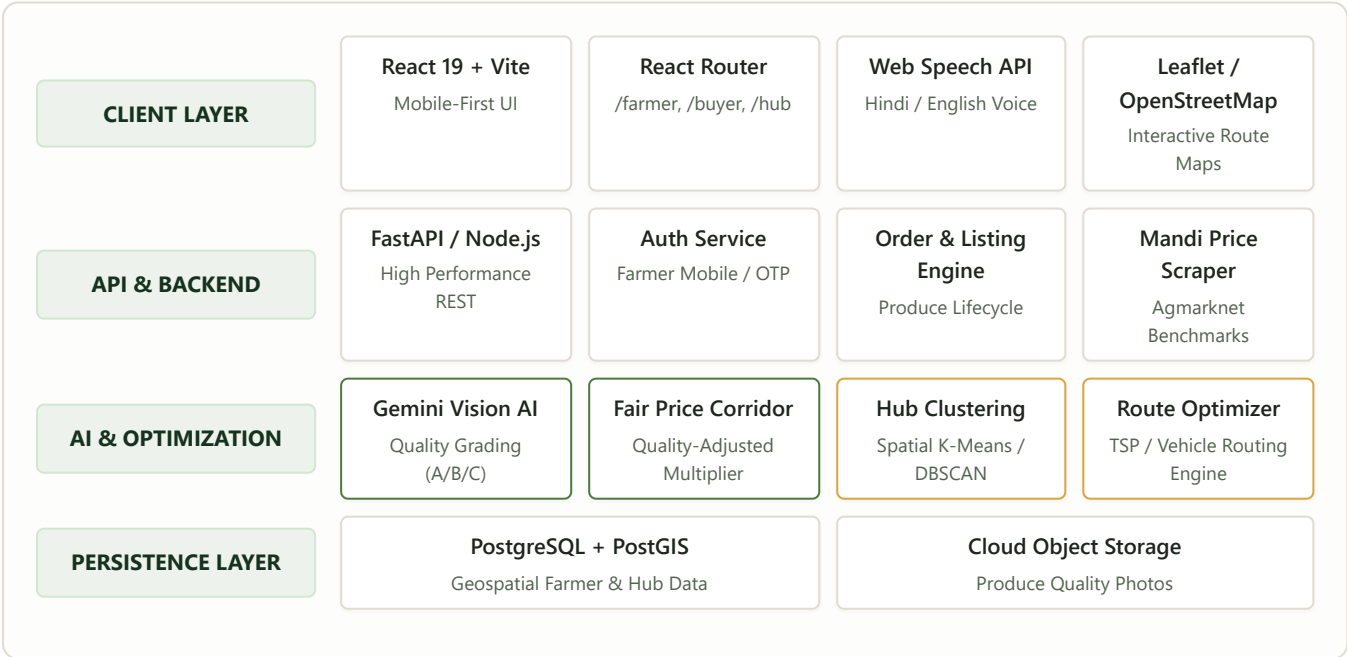
Risk 3: Single-Page Architecture Lock: Currently, all views are jammed into `App.jsx` with modal overlays. Attempting to add multi-step farmer forms, image capture, catalog browsing, and maps within modals will result in fragile, unmaintainable spaghetti code. A dedicated routing system (`react-router-dom`) is mandatory.

Risk 4: React 19 Peer Dependency Compatibility: When installing ecosystem packages (e.g. mapping libraries, chart libraries, or specialized UI widgets), certain npm packages may strictly enforce React 18 in their `peerDependencies`. Packages must be selected for React 19 compatibility or installed with appropriate peer resolution.

Risk 5: AI Workload Placement (Client vs. Server): Running computer vision models directly in rural client mobile browsers causes device crashes and exposes private API keys. AI processing (Gemini Vision / price models) must reside in a secure backend microservice.

6 Recommended Target Architecture for MVP

A modular, scalable architecture separating client presentation, REST orchestration, AI inference, and geospatial persistence:



7 Prioritized Implementation Plan (SIH Execution Roadmap)

PHASE 0 Foundation, Git Initialization & Cleanup

Stabilize workspace, resolve lint warnings, and set up source control.

- Initialize Git repository (`git init`), create initial commit, and establish feature branching.
- Split `LanguageContext.jsx` (separate `useLanguage` hook file) to eliminate Oxlint fast-refresh warning.
- Prune dead starter files (`src/App.css`, unreferenced starter SVGs).

PHASE 1 Client Routing & Role-Based Portals

Transition from a static single page into a full multi-view application.

- Install and configure `react-router-dom` with clean layout wrappers.
- Create routes: `/` (Landing Page), `/farmer` (Farmer Portal), `/buyer` (Buyer Marketplace), `/logistics` (Hub & Route Visualizer).
- Wire Hero and User Type CTA buttons to route directly to respective portals.

PHASE 2 Farmer Produce Listing & AI Quality Grading

Allow farmers to photograph crops and receive an instant automated quality score.

- Build mobile-friendly crop listing form (crop type, quantity, harvest timestamp, GPS coordinates).
- Add camera/photo upload handler with image preview and compression.
- Integrate Gemini Multimodal API endpoint to analyze produce defects, color, and size uniformity; output transparent **Grade A, B, or C** rating.

PHASE 3 AI Fair Price Recommendation Engine

Provide price transparency that eliminates middleman price gouging.

- Establish APMC Mandi market price benchmark integration.
- Apply AI quality grade multipliers (+15% Grade A premium, base Grade B, -10% Grade C).
- Display interactive price corridor (e.g., Recommended ₹28 - ₹32/kg vs. traditional middleman payout of ₹18/kg).

PHASE 4 Buyer Marketplace & Direct Matchmaking

Enable commercial and retail buyers to source directly from farmers.

- Build searchable marketplace catalog with filters for crop category, quality grade, distance, and quantity.
- Implement buyer bulk demand posting interface.
- Build transparent order checkout and direct farmer contact flow.

PHASE 5 Smart Hub Aggregation & Logistics Route Optimizer

Solve the rural logistics bottleneck with batching and optimal transit paths.

- Implement spatial clustering to group nearby farmer listings into designated collection hubs.
- Embed interactive Leaflet / OpenStreetMap visualizer displaying collection hubs, farm pickups, and buyer destinations.
- Solve Multi-Stop Vehicle Routing Problem (VRP) to display the optimal, fuel-efficient delivery trajectory.

PHASE 6 Multilingual Voice Assistant & Hackathon Polish

Empower non-tech-savvy farmers with voice-driven operations.

- Integrate Web Speech API (`SpeechRecognition`) supporting Hindi and English speech inputs.
- Parse spoken queries (e.g., "मुझे 25 किंटल गेहूं बेचना है") directly into auto-populated listing inputs.
- Conduct end-to-end rehearsal with mock rural network bandwidth simulation.